REMARK profile: GEO validation cohorts

Cohort	Step	Patients
GSE65858	GSE65858 series samples	270
GSE65858	HPV16 DNA−negative (hpv16_dna_rna = DNA−)	196
GSE65858	and primary tumour (tumor_type = Primary)	179
GSE65858	analysed for PFS (time > 0, event known)	179
GSE65858	complete covariates for the PFS multivariable model	176
GSE65858	analysed for OS (time > 0, event known)	179
GSE65858	complete covariates for the OS multivariable model	176
GSE41613	GSE41613 series samples	97
GSE41613	HPV−negative OSCC (tissue type)	97
GSE41613	analysed for DSS (time > 0, event known)	97
GSE41613	complete covariates for the DSS multivariable model	97
GSE41613	analysed for OS (time > 0, event known)	97
GSE41613	complete covariates for the OS multivariable model	97
GSE41613	analysed for DSS_unk (time > 0, event known)	97
GSE41613	complete covariates for the DSS_unk multivariable model	97

Cohort	Endpoint	Patients	Events	Median FU (months)	Age	Stage	Site	Missing
GSE65858	PFS (primary)	179	86	32.3	median 57.9 (35.3−87.4)	I−II 41; III 24; IV 114	oral cavity 65; larynx 40; other 71; NA 3	age 0; stage 0; site 3
GSE65858	OS (secondary)	179	63	32.3	median 57.9 (35.3−87.4)	I−II 41; III 24; IV 114	oral cavity 65; larynx 40; other 71; NA 3	age 0; stage 0; site 3
GSE41613	DSS (primary)	97	30	61.7	19−39 6; 40−49 16; 50−59 28; 60−88 47	I/II 41; III/IV 56	oral cavity 97	age 0; stage 0
GSE41613	OS (secondary)	97	51	66.6	19−39 6; 40−49 16; 50−59 28; 60−88 47	I/II 41; III/IV 56	oral cavity 97	age 0; stage 0
GSE41613	DSS_unk (sensitivity)	97	37	62.9	19−39 6; 40−49 16; 50−59 28; 60−88 47	I/II 41; III/IV 56	oral cavity 97	age 0; stage 0

Cohort	Set	Endpoint	Univariable HR (95% CI)	Multivariable HR (95% CI)	PH test p (score)
GSE41613	Buffa 2010 hypoxia	DSS	2.21 (1.45−3.36)	2.10 (1.37−3.23)	0.3
GSE41613	Gavish MP1 G2/M	DSS	1.75 (1.15−2.67)	1.79 (1.18−2.72)	0.58
GSE41613	Gavish MP2 G1/S	DSS	1.45 (1.01−2.09)	1.62 (1.09−2.40)	0.41
GSE41613	Puram 2017 p−EMT	DSS	1.91 (1.31−2.77)	1.97 (1.31−2.97)	0.79
GSE41613	Toustrup 2011 hypoxia	DSS	1.67 (1.16−2.41)	1.63 (1.11−2.39)	0.45
GSE65858	Buffa 2010 hypoxia	PFS	0.93 (0.75−1.15)	0.93 (0.74−1.17)	0.71
GSE65858	Gavish MP1 G2/M	PFS	0.93 (0.75−1.15)	0.88 (0.71−1.09)	0.26
GSE65858	Gavish MP2 G1/S	PFS	0.92 (0.74−1.14)	0.85 (0.68−1.06)	0.27
GSE65858	Puram 2017 p−EMT	PFS	0.99 (0.79−1.23)	0.99 (0.79−1.25)	0.66
GSE65858	Toustrup 2011 hypoxia	PFS	0.97 (0.79−1.20)	1.02 (0.82−1.28)	0.43

Cut−off rule: median of each score in the analysed patients (pre−specified; no optimal cut−point search).
Missing covariates: complete case per model (GSE65858: 3 patients without site are dropped from multivariable models).
GSE65858 times in days (assumed; converted to months for display); GSE41613 times in months (Lohavanichbutr 2013).
GSE41613 age is available only in 4 groups, entered as an ordinal score; DSS_unk = unknown−cause deaths counted as events.